MSG No. 21.39 $C2221'$ [Type II, orthorhombic]

Table 1: Wyckoff site: 2a, site symmetry: 2221'

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5,6,7,8,13,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 2221'

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2}, 0, 0\right]$	[5,6,7,8,13,14,15,16]

Table 3: Wyckoff site: 2c, site symmetry: 2221'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,7,8,13,14,15,16]

Table 4: Wyckoff site: 2d, site symmetry: 2221'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,7,8,13,14,15,16]

Table 5: Wyckoff site: 4e, site symmetry: 2..1'

No.	position	mapping
1	$[x, 0, 0]$	[1,2,9,10]
2	$[-x, 0, 0]$	[3,4,11,12]
3	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	[5,6,13,14]
4	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	[7,8,15,16]

Table 6: Wyckoff site: 4f, site symmetry: 2..1'

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,2,9,10]
2	$\left[-x, 0, \frac{1}{2}\right]$	[3,4,11,12]
3	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,13,14]
4	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	[7,8,15,16]

Table 7: Wyckoff site: `4g`, site symmetry: `.2.1'`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3,9,11]`
2	$[0,\ -y,\ 0]$	`[2,4,10,12]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[6,8,14,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,3,9,11]`
2	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,4,10,12]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,8,14,16]`

Table 9: Wyckoff site: `4i`, site symmetry: `..21'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,9,12]`
2	$[0,\ 0,\ -z]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[6,7,14,15]`

Table 10: Wyckoff site: `4j`, site symmetry: `..21'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,9,12]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ 0,\ z\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[6,7,14,15]`

Table 11: Wyckoff site: `4k`, site symmetry: `..21'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,8,9,16]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[2,7,10,15]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[3,6,11,14]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[4,5,12,13]`

Table 12: Wyckoff site: `8l`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,9]`

continued ...

Table 12

No.	position	mapping
2	$[x, -y, -z]$	`[2,10]`
3	$[-x, y, -z]$	`[3,11]`
4	$[-x, -y, z]$	`[4,12]`
5	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$	`[5,13]`
6	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	`[6,14]`
7	$\left[\frac{1}{2}-x, y+\frac{1}{2}, -z\right]$	`[7,15]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	`[8,16]`